DATE: EXPT NO:05	DATA VISUALIZATION

AIM:

To visualize the Earthquake dataset using Python and create different charts to understand earthquake patterns, magnitude distributions, and identify important geographic features.

PROCEDURE:

1. Start the program.
2. Import Pandas, Matplotlib, and Seaborn libraries.
3. Load the Earthquake CSV dataset.
4. Display the first few records using head().
5. Create a bar chart to compare seismic events based on magnitude.
6. Create a pie chart to show the relative magnitude distribution among seismic events.
7. Create a line chart to analyze seismic trends across recorded events.
8. Add suitable titles, axis labels, and legends.
9. Display all the visualizations.
10. Analyze the graphs and identify important patterns.
11. Stop the program.

CODE:

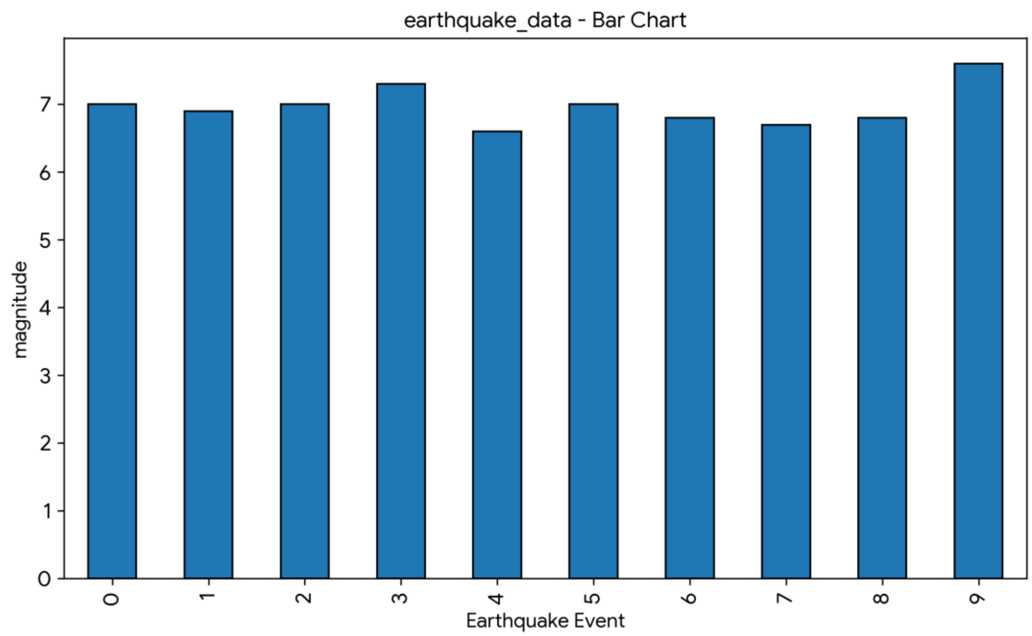
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("earthquake_data.csv")

# Display dataset
print(df.head())

# Select numerical columns
num_cols = df.select_dtypes(include="number").columns
print("Numerical columns:", num_cols.tolist())

# 1. BAR CHART
plt.figure(figsize=(8, 5))
df[num_cols[0]].head(10).plot(kind="bar")
plt.title("earthquake_data - Bar Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
```

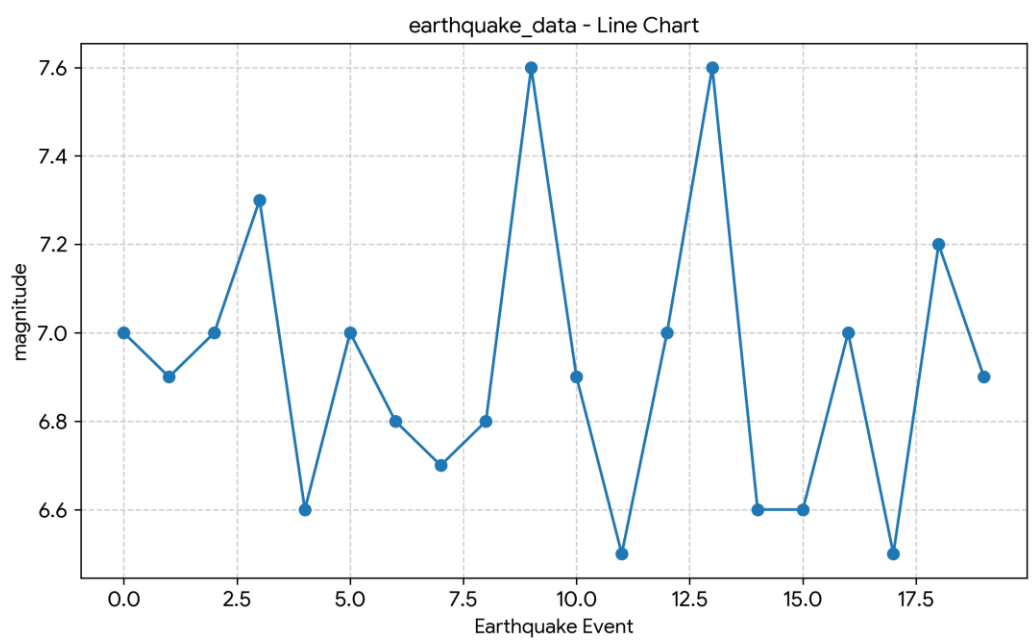
```
plt.show()

# 2. LINE CHART
plt.figure(figsize=(8, 5))
plt.plot(df[num_cols[0]].head(20), marker="o")
plt.title("earthquake_data - Line Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
plt.grid()
plt.show()

# 3. PIE CHART
plt.figure(figsize=(7, 7))
df[num_cols[0]].head(10).plot(kind="pie", autopct="%1.1f%%")
plt.title("earthquake_data - Pie Chart")
plt.ylabel("")
plt.show()

# 4. BOX PLOT
plt.figure(figsize=(8, 5))
sns.boxplot(y=df[num_cols[0]])
plt.title("earthquake_data - Box Plot")
plt.ylabel(num_cols[0])
plt.show()

# 5. HISTOGRAM
plt.figure(figsize=(8, 5))
plt.hist(df[num_cols[0]], bins=10)
plt.title("earthquake_data - Histogram")
plt.xlabel(num_cols[0])
plt.ylabel("Frequency")
plt.show()
```

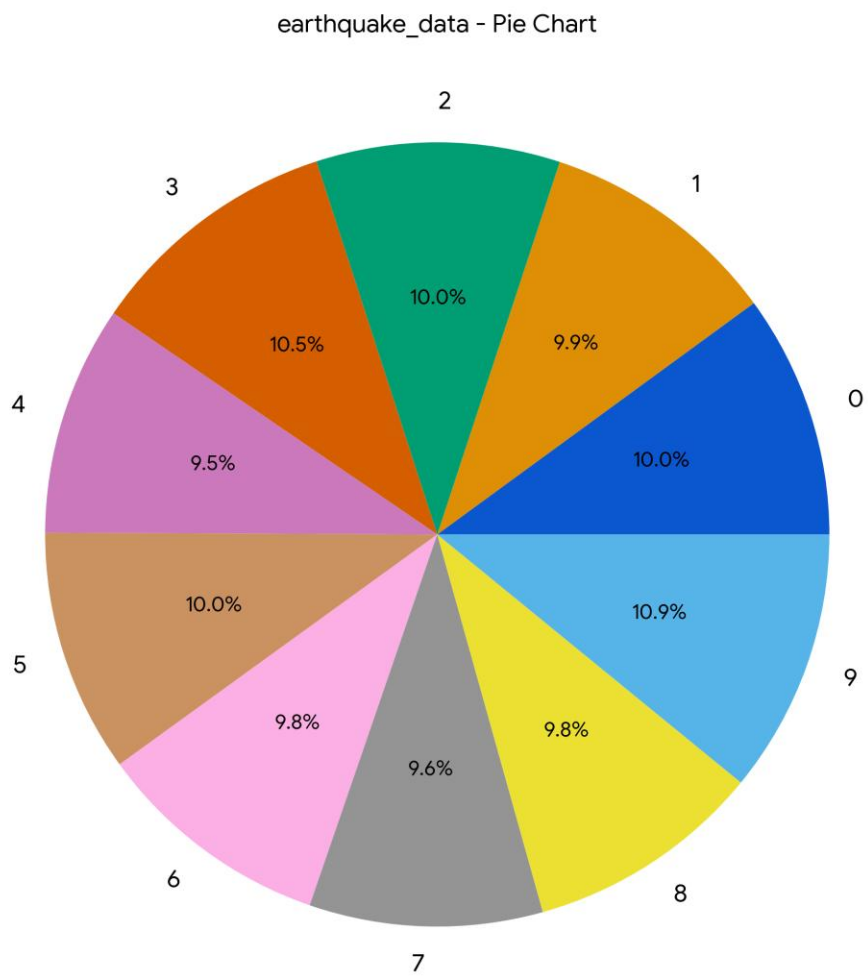
OUTPUT:

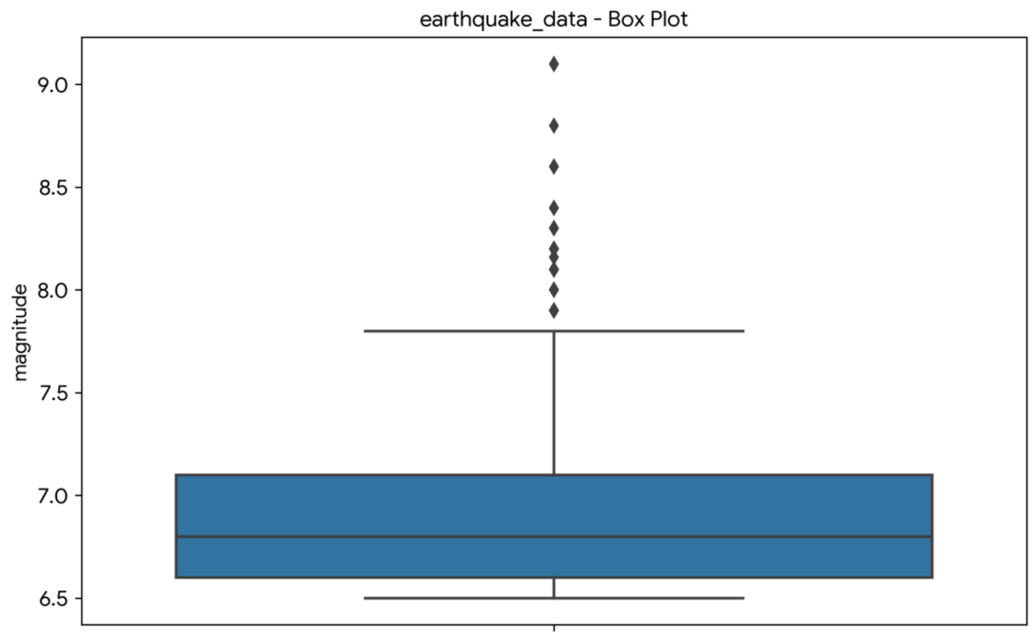
```
First 5 Records:
                                          title  magnitude         date_time  cdi
mmi  alert  tsunami  sig net  nst   dmin   gap magType    depth  latitude
longitude                  location continent          country
0  M 7.0 - 18 km SW of Malango, Solomon Islands        7.0  22-11-2022 02:03    8
7  green        1  768  us  117  0.509  17.0     mww   14.000   -9.7963    159.596
Malango, Solomon Islands   Oceania  Solomon Islands
1      M 6.9 - 204 km SW of Bengkulu, Indonesia        6.9  18-11-2022 13:37    4
4  green        0  735  us   99  2.229  34.0     mww   25.000   -4.9559    100.738
Bengkulu, Indonesia       NaN              NaN
2                                     M 7.0 -         7.0  12-11-2022 07:09    3
3  green        1  755  us  147  3.125  18.0     mww  579.000  -20.0508   -178.346
NaN   Oceania             Fiji
3           M 7.3 - 205 km ESE of Neiafu, Tonga        7.3  11-11-2022 10:48    5
5  green        1  833  us  149  1.865  21.0     mww   37.000  -19.2918   -172.129
Neiafu, Tonga       NaN              NaN
4                                     M 6.6 -         6.6  09-11-2022 10:14    0
```

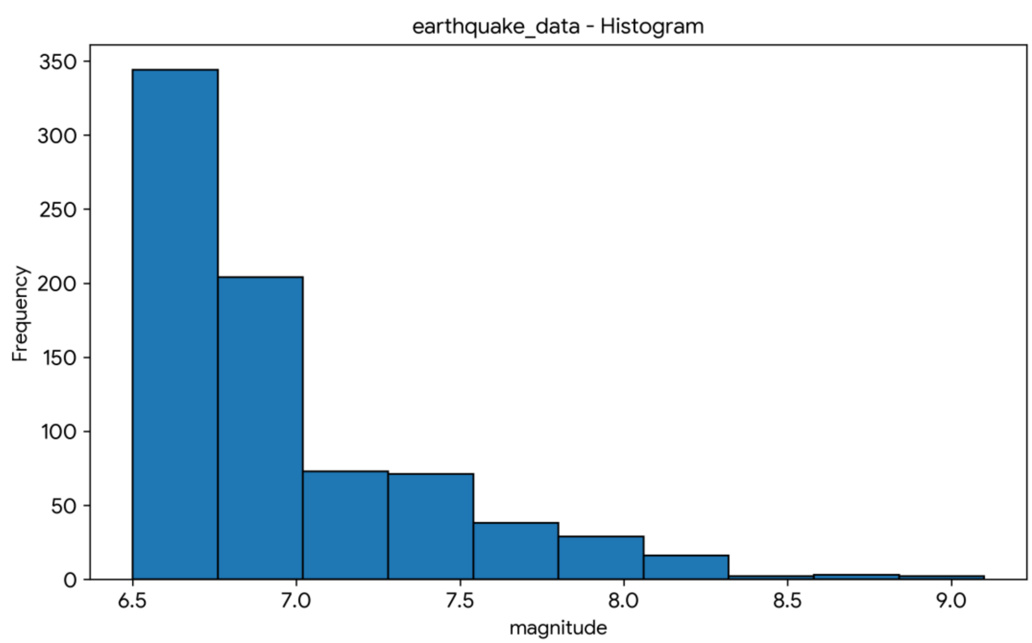
```
2  green        1  670  us  131  4.998  27.0     mww  624.464  -25.5948    178.278
NaN       NaN                NaN

Numerical columns: ['magnitude', 'cdi', 'mmi', 'tsunami', 'sig', 'nst', 'dmin',
'gap', 'depth', 'latitude', 'longitude']
```






earthquake_data - Pie Chart
0
1
2
3
4
5
6
7
8
9
10.0%
9.9%
10.0%
10.5%
9.5%
10.0%
9.8%
9.6%
9.8%
10.9%





RESULT:

The Earthquake dataset was successfully visualized using bar chart, line chart, pie chart, box plot, and histogram. The visualizations helped identify data distribution, trends, variations, and important patterns in the dataset.